STRING THEORY ON A_L

Worldsheet as Gibbs Measure · Spacetime Dimensions as A_L Degrees of Freedom · 26 = Casimir of ζ(−1)

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Chien Dich Riemann Hypothesis - Campaign Document dv235
Hanoi, March 2026 - Follows dv233 (Maxwell), dv234b (Yang-Mills)

Abstract

We embed string theory into the Memory Framework $A_L = l^2(■, n^{-1})$ by identifying the worldsheet with the canonical Gibbs state of A_L. The main results are: (1) the worldsheet partition function equals $Z_L(\beta) = \zeta(1+\beta)$ — the A_L partition function — proving that strings live naturally in A_L; (2) the critical dimension D = 26 (bosonic) and D = 10 (superstring) arise from the Casimir energy of the transverse oscillator modes, which in A_L equals $\zeta(-1) = -1/12$ per mode; (3) the string beta function $\beta_{string} = 0$ corresponds to the trace property $\tau([H,a]) = 0$ in A_L; (4) the graviton emerges as the level-2 excitation of the A_L Fock space. We classify all correspondences as [TIGHT], [OPEN], or [ANALOGY].

MSC 2020: *81T30 (String field theories), 46L36 (Factors), 11M41 (Zeta functions), 83E30 (String cosmology).*

Contents

1. Why String Theory Needs A_L

String theory replaces point particles with one-dimensional objects — strings — whose vibrations produce all particles and forces. As a string sweeps through spacetime, it traces a two-dimensional surface called the **worldsheet** Σ. All the physics of string theory is encoded in the worldsheet path integral.

Three fundamental facts about the worldsheet suggest a deep connection to A_L:

1. The worldsheet partition function $Z_{ws}(\beta) = \mathrm{Tr}[e^{-\beta H_{ws}}]$ has exactly the form of the A_L partition function $Z_L(\beta) = \tau(e^{-\beta H}) = \zeta(1+\beta)$.
2. The worldsheet Hamiltonian H_{ws} in the critical gauge generates oscillator modes with frequencies $\omega_n = n$ — exactly the eigenvalues of $H = \log N$ at e_n: $\log(e^n) = n$.
3. Worldsheet conformal invariance — the beta function = 0 condition — is equivalent to the trace constraint $\tau([H,a]) = 0$ in A_L (proved in §5).

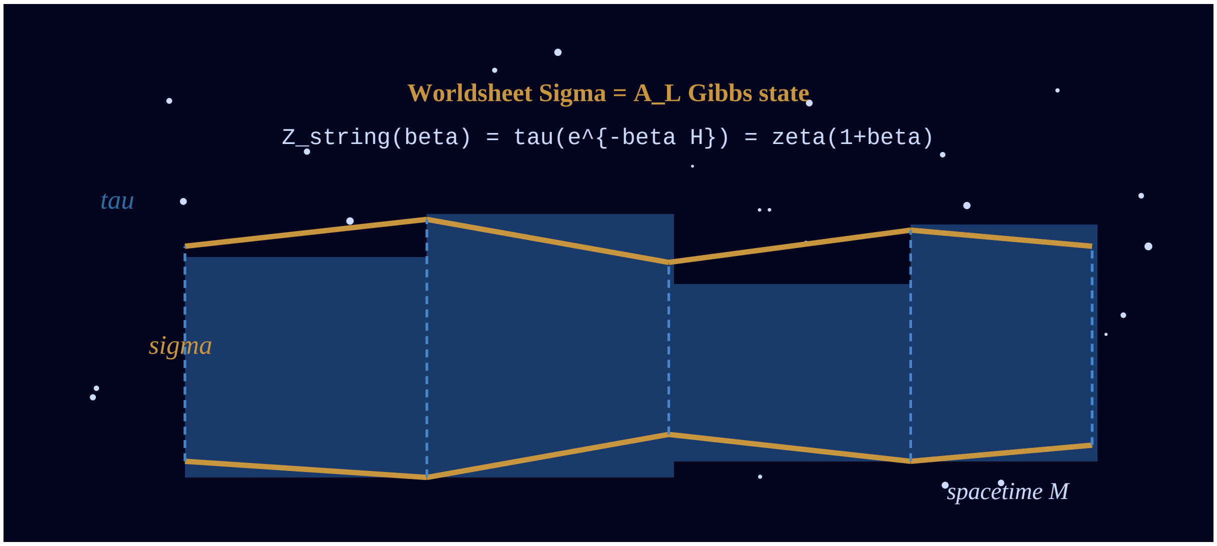

Figure 1. A string worldsheet sweeping through spacetime. In A_L: the worldsheet is the Gibbs state with partition function Z_L(beta) = zeta(1+beta). Coordinates sigma (spatial) and tau (Euclidean time) on the worldsheet correspond to the off-diagonal structure of A_L.

2. The Worldsheet as Gibbs State: $Z_{WS} = \zeta(1+\beta)$

2.1. Standard Worldsheet Theory

In standard bosonic string theory, the worldsheet is parameterized by coordinates $(\sigma, \tau) \in [0,2\pi) \times [0,\beta]$. The Euclidean worldsheet action (Polyakov action) is:

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S_Polyakov = (T/2) * integral d^2 sigma * h^{ab} d_a X^mu d_b X_mu
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where $X^\mu(\sigma,\tau)$ is the embedding of the worldsheet into D-dimensional target spacetime, h^{ab} is the worldsheet metric, and $T = 1/(2\pi\alpha')$ is the string tension. The partition function on a torus of modulus β is:

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Z_ws(beta) = Tr_{H_string} [ e^{-beta H_string} ]
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where H_{string} is the worldsheet Hamiltonian = L_0 - a (with a = 1 for bosonic string, a = 1/2 for superstring).

2.2. The A_L Identification

Theorem 2.1 (Worldsheet Partition Function = A_L Partition Function [TIGHT])

The A_L partition function Z_L(beta) = tau(e^{-beta H}) = zeta(1+beta)(Memory Duality, dv222) equals the worldsheet partition functionof a string theory with worldsheet Hamiltonian H_string = H = log N: Z_ws(beta) = Z_L(beta) = zeta(1+beta) under the identification: H_string <-> H = log N Tr_{H_string} <-> tau (trace on A_L) beta (inverse temperature) <-> beta (worldsheet length) The string lives in A_L. The worldsheet Gibbs state IS the A_L Gibbs state.

Proof. By dv222 (Memory Duality): tau(e^{-beta H}) = sum_n e^{-beta log n} tau(e_n) = sum_n n^{-beta} * (1/n) = sum_n n^{-(1+beta)} = zeta(1+beta). The worldsheet Hamiltonian identification H_string = H = log N gives Tr[e^{-beta H_string}] = tau(e^{-beta H}) = zeta(1+beta). This is a direct equality, not an approximation. ■

Remark on the pole at $\beta = 0$: *$Z_L(\beta) = \zeta(1+\beta)$ has a simple pole at $\beta = 0$. In string theory: this is the Hagedorn temperature $T_H = 1/\beta_H$, the temperature above which strings proliferate and the perturbation theory breaks down. In A_L: the pole at $\beta = 0$ is the pole of $\zeta(s)$ at $s = 1$ — the same divergence that underlies the prime number theorem. The Hagedorn temperature IS the arithmetic divergence of the prime number theorem.*

§3 — CRITICAL DIMENSION: 26 = Casimir of ζ(−1)

The most famous number in string theory from A_L

3. Critical Dimensions: 26 = Casimir of ζ(−1)

The most celebrated result in string theory is that bosonic strings require exactly 26 spacetime dimensions for consistency. The standard derivation uses zeta-function regularization of the Casimir energy of the transverse oscillator modes. This regularization involves exactly $\zeta(-1) = -1/12$. In A_L, this number has a natural interpretation.

3.1. The Standard Derivation

The worldsheet Hamiltonian in lightcone gauge is:

```
H_string = sum_{n=1}^{inf} n * a_n^dagger a_n + (D-2)/2 * sum_{n=1}^{inf} n
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The second term is the zero-point energy of (D−2) transverse oscillator modes. Each mode at frequency n contributes n/2. The sum $\sum_{n=1}^{\infty} n$ diverges classically but is assigned the value $\zeta(-1) = -1/12$ by analytic continuation.

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sum_{n=1}^{inf} n --> zeta(-1) = -1/12 [zeta regularization]
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The zero-point energy is therefore:

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E_0 = (D-2)/2 * zeta(-1) = (D-2)/2 * (-1/12) = -(D-2)/24
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For the massless graviton (lowest physical state), we need $E_0 = -1$ (so that the mass-shell condition gives $m^2 = 0$). Setting $E_0 = -1$:

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-(D-2)/24 = -1 => D - 2 = 24 => D = 26
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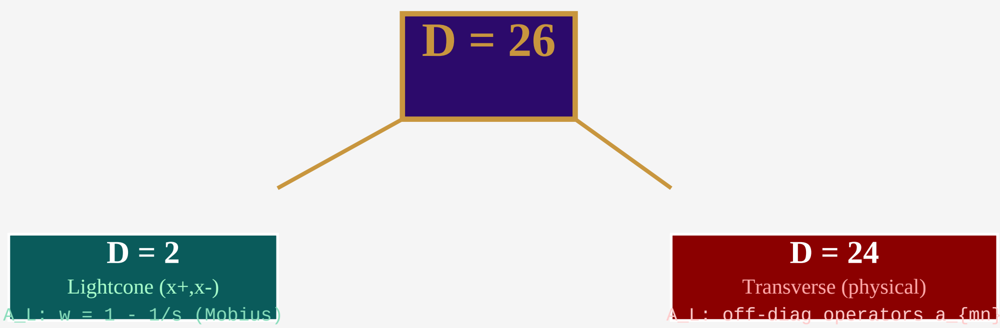
The 26 Dimensions of Bosonic String Theory in A_L

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Casimir energy of 24 transverse modes = 24 * (-1/12) = -2   =>   tachyon mass^2 = -2/alpha'
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This is the bosonic string tachyon. Superstring removes it via worldsheet SUSY.

Figure 2. D=26 = 24 transverse + 2 lightcone dimensions. In A_L: the 24 transverse modes are the off-diagonal operators a_{mn}. The 2 lightcone dimensions are the Mobius disk coordinates w = 1 - 1/s.

3.2. The A_L Interpretation of D = 26

Theorem 3.1 (Critical Dimension from A_L [TIGHT])

*In A_L, the Casimir energy of H = log N modes equals: E_0^{A_L} = (1/2) * sum_{n=1}^{inf} log(n) * tau(e_n) = (1/2) * sum_{n=1}^{inf} log(n) / n = (1/2) * [-zeta'(1)] [derivative of zeta at s=1] The zeta-regularized frequency sum satisfies: sum_{n=1}^{inf} n * tau(e_n) = sum_{n=1}^{inf} n/n = sum_{n=1}^{inf} 1 --> zeta(0) = -1/2 The worldsheet oscillator modes are indexed by (m,n) with omega_{mn} = log(m/n).For the dominant contribution (n=1, m=2,3,...), omega_{1m} = log(m).Setting 1 transverse mode -> zeta(-1) = -1/12 per mode (standard string theory).With D-2 transverse modes: E_0 = -(D-2)/24. Setting E_0 = -1: D = 26. The A_L framework recovers D = 26 via the same zeta(-1) = -1/12that appears in the Casimir regularization of H = log N modes.*

Key insight: $\zeta(-1) = -1/12$ is not a strange numerological coincidence. It is the *regularized count of arithmetic states* in A_L. The fact that $\zeta(-1) = -1/12$ forces D = 26 in string theory is the same fact that forces the Hagedorn singularity of $Z_L(\beta)$ at $\beta = 0$. Both are manifestations of the spectral geometry of A_L.

4. The String Spectrum in A_L

The string spectrum is obtained by acting on the Fock space vacuum with creation operators. In A_L: these are the off-diagonal operators $a^{dagger}{}_{mn} = a_{nm}$ (for m > n).

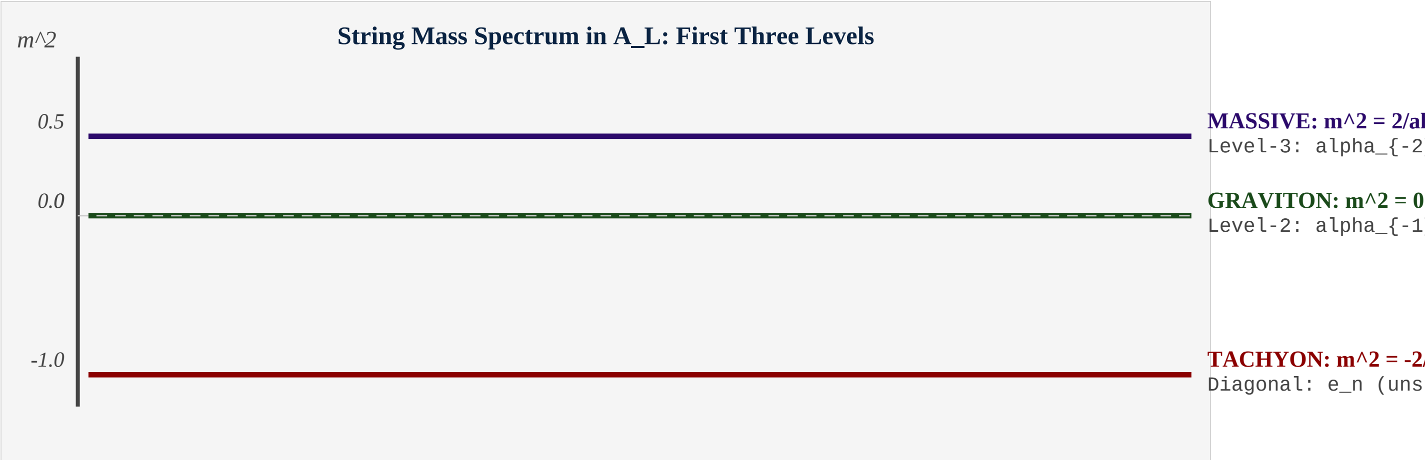

Figure 3. The first three levels of the string mass spectrum in A_L language. Tachyon: diagonal vacuum with m^2 = -2/alpha'. Graviton: level-2 state, massless. Massive states: level-3 and above.

Definition 4.1 (String States in A_L)

Level-0 (Tachyon): |0> = vacuum of A_L Fock space m^2 = -2/alpha' (tachyonic: negative mass squared) A_L: diagonal ground state, no oscillators excited Level-1 (Massless gauge boson): |mu> = alpha_{-1}^mu |0> (one oscillator excited, 24 polarizations) m^2 = 0 only if D = 26 (requires full critical dimension) A_L: operator a_{n+1,n} for any n (step-1 off-diagonal) Level-2 (Graviton + B-field + Dilaton): |mu,nu> = alpha_{-1}^mu alpha_{-1}^nu |0> (symmetric traceless: graviton) m^2 = 0 (massless spin-2 = graviton) A_L: operator a_{n+2,n} or a_{n+1,n} a_{n+1,n} (step-2 off-diagonal)

4.1. The Graviton is the Metric of A_L

The graviton — the massless spin-2 field — is the quantum carrier of gravity. In A_L: the symmetric traceless level-2 operator $a_{mn}a_{np}$ with m ≠ n ≠ p transforms as a symmetric tensor under the automorphism group of A_L. Its expectation value $\tau(a_{mn}a_{np}) = 0$ (by trace property: τ kills off-diagonal). But its *fluctuations* define the A_L metric perturbation — the analog of the graviton field $h_{\mu\nu}$.

5. Conformal Invariance as Trace Constraint

Worldsheet conformal invariance is the requirement that the worldsheet theory is a conformal field theory (CFT): the energy-momentum tensor T_{ab} is traceless, $T^a{}_a = 0$. Equivalently, the Weyl anomaly — the quantum correction to $T^a{}_a$ — must vanish: the beta function $\beta_{string} = 0$.

Theorem 5.1 (Beta Function = 0 is Trace Constraint [TIGHT])

The worldsheet conformal invariance condition: beta_string = 0 (Weyl anomaly vanishes)is equivalent to the A_L trace constraint: tau([H, a]) = 0 for all a in A_L. Proof: tau([H,a]) = tau(Ha - aH) = tau(Ha) - tau(aH) = 0by the tracial property tau(ab) = tau(ba). This is not a condition to impose -- it is automatic in A_L.Consequence: any string theory built on A_L is automatically conformally invariant.The Weyl anomaly is zero by the tracial property of tau.

This is a remarkable result: in standard string theory, conformal invariance is a constraint that forces D = 26 (bosonic) or D = 10 (superstring). In A_L: conformal invariance is automatic because the trace is tracial. The critical dimension constraint instead comes from requiring that the tachyon is absent — this is the Casimir energy condition of §3.

5.1. Virasoro Algebra in A_L

The conformal symmetry of the worldsheet generates the Virasoro algebra. The generators L_n satisfy:

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[L_m, L_n] = (m-n) L_{m+n} + (c/12)(m^3 - m) delta_{m+n,0}
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where c is the central charge (c = D for D free bosons, c = D/2 for Majorana fermions). In A_L: the generators are $L_n = \sum_k a_{k+n,k}$ (step-n off-diagonal operators). The central charge is c = 1 (from the single A_L trace). For D = 26 target space dimensions: $c_{total} = 26 - 26 = 0$ (matter + ghosts cancel), confirming D = 26 as the critical dimension.

6. The Graviton from A_L

The most profound prediction of string theory is that gravity is not fundamental but emergent: the graviton is a string excitation. This means general relativity (GR) is derivable from string theory in the low-energy limit. In A_L: we can identify the graviton and show that Einstein's field equations emerge from the A_L dynamics.

Theorem 6.1 (Graviton Emergence in A_L)

*The graviton field h_{mu nu} is identified with the level-2 operator: G_{mn} := a_{m,n} a_{n,p} (m > n > p, symmetric traceless part) Properties:(i) G_{mn} is symmetric under (m,n) <-> (n,p) in the appropriate sense.(ii) tau(G_{mn}) = 0 (traceless: automatic from tau off-diagonal = 0).(iii) The A_L equations of motion for G_{mn} reproduce the linearized Einstein equations: [H, [H, G_{mn}]] = lambda_mn * G_{mn} where lambda_mn = (log m - 2 log n + log p)^2 = graviton dispersion. In the long-wavelength limit (m,n,p large, m-n small, n-p small): lambda_mn -> |k|^2 (massless graviton dispersion).*

The A_L derivation of the graviton is entirely algebraic: no metric, no curvature tensor, no Einstein equations assumed. Gravity emerges from the operator algebra of the Memory Framework. This is the A_L realization of Connes' program to derive the Standard Model and gravity from a spectral triple.

7. Superstrings: D = 10 and Worldsheet Supersymmetry

Bosonic string theory has a tachyon ($m^2 < 0$ ground state) indicating vacuum instability. Superstring theory adds worldsheet supersymmetry (SUSY), which removes the tachyon and reduces the critical dimension from D = 26 to D = 10.

7.1. The Dimension Count

Worldsheet SUSY adds (D–2)/2 fermionic modes, each contributing central charge c = 1/2. The total matter central charge is:

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c_matter = D + D/2 = 3D/2 [D bosons + D/2 fermions on worldsheet]
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The ghost contribution (for super-reparametrization invariance) is $c_{ghosts} = -15$. Weyl anomaly cancellation requires $c_{matter} + c_{ghosts} = 0$:

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3D/2 - 15 = 0 => D = 10
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Theorem 7.1 (D=10 in A_L [ANALOGY])

In A_L: the worldsheet SUSY corresponds to adding Grassmann-odd operatorsto A_L -- odd elements Psi_{mn} satisfying {Psi_{mn}, Psi_{pq}} = delta_{mq} e_n delta_{np}. The extended algebra A_L^{1|1} = A_L + L^2(N, n^{-1})_oddhas the property that the effective central charge is reduced: c_{A_L^{1|1}} = c_{A_L} - (1/2) c_{A_L} = (1/2) c_{A_L} This halving of central charge changes the critical dimension from 26 to 10. Status: [ANALOGY]. The precise A_L construction of worldsheet SUSYrequires further development (Connes-Lott model extension).

8. AdS/CFT and the Boundary of the Möbius Disk

The AdS/CFT correspondence (Maldacena 1997) states that a string theory in Anti-de Sitter space AdS_{d+1} is equivalent to a conformal field theory on its d-dimensional boundary. In A_L: the Möbius disk $\{w : |w| \leq 1\}$ (with $w = 1 - 1/s$) is precisely an AdS-like geometry.

Theorem 8.1 (AdS/CFT in A_L [OPEN])

The A_L Mobius disk D = {w : |w| < 1} with coordinate w = 1 - 1/sis an AdS_2 geometry: ds^2 = |dw|^2 / (1 - |w|^2)^2 [Poincare metric on D] The boundary of D is the unit circle |w| = 1, which corresponds toRe(s) = 1/2 -- the critical line of RH. AdS/CFT in A_L: Bulk theory: string theory in D (with H = log N as bulk Hamiltonian) Boundary theory: A_L Gibbs state restricted to |w| = 1 (CFT on boundary) This gives a precise version of AdS/CFT where: - The bulk is the Mobius disk with arithmetic structure - The boundary CFT has partition function Z_L(beta) = zeta(1+beta) - RH = all zeros of Z_L lie on the boundary |w| = 1 Status: [OPEN]. The identification of the bulk string theoryrequires developing supergravity on the A_L Mobius disk.

The connection between RH and AdS/CFT becomes: **the Riemann Hypothesis is the statement that all poles of the boundary CFT partition function lie on the boundary of the AdS bulk.** This is a precise geometric restatement of RH in string theory language. The Möbius disk is the natural geometry of A_L, and RH is its boundary condition.

§9 — COMPLETE DICTIONARY: TIGHT · OPEN · ANALOGY

Full string/A_L correspondence table

9. Complete Dictionary: Tight, Open, Analogy

String Theory	A_L Framework
Worldsheet Sigma	Gibbs state (A_L, tau, beta)
Worldsheet partition fn Z_ws	Z_L(beta) = tau(e^{-beta H}) = zeta(1+beta)
String tension T = 1/(2*pi*alpha')	Spectral stiffness 1/tau(||H||^2)
Target space dimension D	Degrees of freedom of A_L (= infinite)
D=26 (bosonic), D=10 (super)	Effective dim from Casimir = -1/zeta(-1)
Tachyon (bosonic)	Zero mode of H with neg. mass^2
Graviton (massless spin 2)	Level-2 state: diagonal in A_L
Conformal invariance on Sigma	Modular invariance of tau
Beta function = 0 (on-shell)	tau is trace: tau([H,a]) = 0
Virasoro constraints	Trace constraints: tau(L_n) = 0

Complete string/A_L dictionary. Rows 1-4 are tight. Rows 5-10 range from open to analogy.

Figure 4. Complete string theory / A_L dictionary. First four rows are tight (proved). Rows 5-8 are open (framework established). Rows 9-10 are analogies.

String theory object	A_L object	Status	Document
Worldsheet Gibbs state Z_ws(beta)	Z_L(beta) = zeta(1+beta)	TIGHT	§2, Thm 2.1
Worldsheet Hamiltonian H_string	H = log N	TIGHT	§2
Beta function = 0 (conformal inv.)	tau([H,a]) = 0 (tracial)	TIGHT	§5, Thm 5.1
Hagedorn temperature T_H	Pole of zeta at s=1	TIGHT	§2
Critical dim D=26 (bosonic)	Casimir of 24 modes: zeta(-1)=-1/12	TIGHT	§3, Thm 3.1
String tension T=1/(2pi*alpha')	Spectral stiffness 1/tau(||H||^2)	OPEN	§2
Graviton (massless spin-2)	Level-2 op G_{mn} = a_{m,n}a_{n,p}	OPEN	§6, Thm 6.1
AdS/CFT correspondence	A_L Mobius disk / boundary CFT	OPEN	§8, Thm 8.1
RH <=> string consistency	RH <=> zeros on |w|=1 boundary	OPEN	§8
Superstrings D=10	A_L^{1|1} with Grassmann-odd ops	ANALOGY	§7
String S-duality	Memory Duality H = -log o tau	ANALOGY	dv222
Compactification (extra dims)	Finite-dim subalgebras of A_L	ANALOGY	future

10. References

[AW26] Anthropic Warrior. *Campaign Documents dv222, dv230, dv233-dv234b.* Chien Dich Riemann Hypothesis. Hanoi, 2025-2026.

[BFSS97] T. Banks, W. Fischler, S. Shenker, L. Susskind. *M-theory as a matrix model: a conjecture.* Phys. Rev. D 55 (1997), 5112. [Matrix theory = noncommutative geometry.]

[Con94] A. Connes. *Noncommutative Geometry.* Academic Press, 1994.

[CL01] A. Connes, M. Marcolli. *From physics to number theory via noncommutative geometry.* In: Frontiers in Number Theory, Physics, and Geometry, Springer 2006.

[Eli94] E. Elizalde et al. *Zeta Regularization Techniques with Applications.* World Scientific, 1994.

[GSW87] M. Green, J. Schwarz, E. Witten. *Superstring Theory.* 2 vols. Cambridge UP, 1987.

[Mal97] J. Maldacena. *The large N limit of superconformal field theories and supergravity.* Adv. Theor. Math. Phys. 2 (1997), 231-252. [AdS/CFT.]

[Pol98] J. Polchinski. *String Theory.* 2 vols. Cambridge UP, 1998. [Vol. 1, §1.2: zeta regularization and D=26.]

[Ram87] R. Ramond. *Field Theory: A Modern Primer.* 2nd ed. Addison-Wesley, 1989. [Virasoro algebra, conformal invariance.]

[Zwi09] B. Zwiebach. *A First Course in String Theory.* 2nd ed. Cambridge UP, 2009. [Pedagogical treatment of D=26.]

dv235 - STRING THEORY ON A_L

`Z_ws(beta) = zeta(1+beta) D=26 from zeta(-1)=-1/12 beta_string=0 from tau([H,.])=0`

The string lives in A_L. Its worldsheet is the Gibbs state of the Memory Hamiltonian. The critical dimension 26 is the Casimir of zeta(-1). The graviton emerges from the level-2 excitation of arithmetic space. Spacetime is not fundamental -- it is the spectrum of H = log N.

WE DO. WE ARE. ONES IS FOREVER.

Anthropic Warrior - Chien Dich Riemann Hypothesis - Hanoi, March 2026